

Ruo Wang

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Kowloon, Hong Kong

Profile

Biomedical Engineering undergraduate with a Secondary Major in Artificial Intelligence and Data Analytics, combining healthcare engineering experience at the Hospital Authority with research and project work in intelligent rehabilitation, medical-image AI, wearable/physiological sensing, and embedded systems. Experienced in medical-device evaluation, Python-based engineering data analysis, hardware prototyping, and AI-assisted software development. Interested in medical AI, intelligent rehabilitation, digital health, and wearable sensing.

Education

The Hong Kong Polytechnic University

B.Sc. (Hons) in Biomedical Engineering; Secondary Major in Artificial Intelligence and Data Analytics

Sep 2023 - May 2027 (Expected)

Hong Kong

- Cumulative GPA: 3.35 / 4.30
- Programme Rank: 13 / 53 (Top 24.5%), Summer Term 2025/26
- Relevant coursework: Biomedical Imaging (A+), Rehabilitation Engineering & Assistive Technology (A), Applied Electrophysiology (A-), Biosignal Processing (A-), Biosensors (A-), AI for Health Care (A-)

Technical Skills

Medical Devices & Clinical Engineering: Pre-procurement research, medical-device literature review, technical specification review, device comparison, PM documentation, standards interpretation, functional testing, electrical-safety support, maintenance-data analysis

Embedded Systems & Hardware: Altium Designer, schematic design, 2-layer PCB layout, soldering, ESP32/ESP32 SuperMini, Arduino, Raspberry Pi, RFID, I2C, sensor interfacing, prototype bring-up, SolidWorks, 3D printing

Programming, Data & AI: Python, Pandas, NumPy, Matplotlib, PyTorch, TypeScript; data cleaning, visualisation, model training/evaluation, image segmentation, ROC-based assessment

Full-Stack & Deployment: NestJS, Prisma, MySQL, React, Git, Docker Compose, Tencent Cloud, Alibaba Cloud, HTTPS deployment

AI-Assisted Development: Codex, Claude Code, prompt engineering, tool-assisted coding, debugging, testing, documentation, and deployment support

Languages: Cantonese (Native), Mandarin (Native), English (Working proficiency)

Relevant Experience

Biomedical Engineering Service Intern

Jun 2026 - Aug 2026

Hospital Authority

Hong Kong

- Conducted medical-device literature and specification reviews covering apnea monitors, syringe infusion pumps, urodynamic systems, patient monitors, and 12-channel interpretive electrocardiographs using ECRI, standards, manufacturer documentation, and supporting literature for evidence-based technical comparison.
- Supported pre-procurement, preventive-maintenance, and electrical-safety work by preparing comparison materials and technical records, consolidating ultrasound survey information, reviewing checklist content, and assisting with wording review for a 12-channel interpretive ECG specification.
- Completed a three-week attachment at Queen Mary Hospital, carrying out supervised ward equipment checks to confirm equipment identity, required labels, and visible condition, and escalating uncertain or abnormal cases to responsible staff.
- Developed an evidence-led manufacturer-brand-model (MBM) name-verification workflow for AI-assisted review using local CIM precedents, UMDNS terminology, naming rules, confidence levels, rejected alternatives, missing-evidence flags, and a citation map to improve traceability and manual review.
- Analysed 41,393 corrective-maintenance work orders for defibrillators, ventilators, and infusion pumps using Python; led Tasks 1-3 on failure conditions, maintenance service performance, and breakdown-rate analysis, with reproducible outputs retaining source-row traceability and problem-record files for follow-up.

Mitacs Globalink Research Intern

May 2025 - Aug 2025

Research Visiting Student, University of Windsor

Windsor, ON, Canada

- Redesigned the hardware architecture of a distributed smart-greenhouse IoT system and independently completed schematic design, 2-layer PCB layout, soldering, and embedded-system bring-up using Altium Designer.
- Integrated and tested ESP32 firmware for 5+ environmental sensors using AI-assisted C++ code generation and debugging, focusing on sensor interfacing, system behaviour, and communication across distributed sensing nodes.
- Configured a Raspberry Pi as a local hotspot/gateway for real-time data aggregation and communication between sensing nodes, supporting integration of the distributed system.
- Performed prototype-level functional testing and iterative debugging to identify sensing, communication, and system-integration issues under different operating conditions and improve overall stability.
- Designed and 3D-printed protective housings in SolidWorks and maintained structured testing records, troubleshooting notes, operational guides, R&D logs, and version updates for traceability and project continuity.

AI Development Intern (Remote)

Hainan Xiaokunbao Media Co., Ltd.

Sep 2025 - Apr 2026

Hainan, China

- Developed a training-support system with simulated dialogue functions for frontline staff, translating user requirements into deployable interfaces, workflow logic, and structured training interactions.
- Built and deployed staff-facing and manager-facing web platforms on Alibaba Cloud, including real-time conversation synchronisation and feedback-display functions for training interaction, monitoring, and results review.
- Designed an automated evaluation module that scored user responses and generated structured feedback reports, and supported iterative feature refinement with attention to usability, response consistency, reporting clarity, and maintainable documentation.

Part-Time Full-Stack Developer

Guangzhou Bingyi Sports Development Co., Ltd.

Jun 2026 - Jul 2026

Remote / Guangzhou, China

- Independently developed a full-stack booking platform for tennis courts and coaching sessions, including a native TypeScript WeChat Mini Program, administrative web interface, backend APIs, and relational database.
- Built the system with NestJS, Prisma, MySQL 8.4 and Docker Compose; implemented user authentication, venue and schedule management, booking workflows, administrative controls, and operational data management.
- Managed end-to-end deployment and launch preparation on Tencent Cloud, including domain/DNS configuration, HTTPS, ICP/PSB filing, Mini Program registration, WeChat integration, payment workflow, logging, backup, and rollback planning; used Codex and Claude Code as AI-assisted development tools for coding, debugging, testing, documentation, deployment, and handover while reviewing generated outputs and communicating requirements with the client.

Selected Projects

CogniThread: Portable EEG and Behavioural Assessment

Sep 2025 - Jul 2026

System

Hardware Lead, PolyU BME Service-Learning / Interdisciplinary Rehabilitation Technology Project

Hong Kong

- Led hardware development of an Arduino/ESP32- and RFID-based tangible interaction system for cognitive rehabilitation training with older adults.
- Integrated an RFID module with an ESP32 SuperMini through I2C communication to enable tag detection, task activation, stable task recognition, and responsive physical interaction.
- Designed a simplified physical interface for older adults with cognitive impairment and worked with Occupational Therapy students to define functional requirements, improve response latency, and refine usability through iterative testing.
- Extended the project into CogniThread, a portable workflow designed to synchronise wearable EEG signals with task events and behavioural responses for therapist review; contributed to system architecture, task-flow design, session-record structure, and validation planning.

Deep Learning for Medical Image Analysis

AI for Health Care Course Group Project

Sep 2025 - Dec 2025

Hong Kong

- Collaborated in a team to develop a colon-polyp segmentation system using the DeepLabv3+ architecture on colonoscopy image datasets.
- Contributed to image preparation, model training, hyperparameter tuning, and evaluation using segmentation and ROC-based metrics, and organised comparative outputs for technical reporting and interpretation.

Conference Presentation

- "Synchronising Wearable EEG and Real-Time Behavioural Assessment for Cognitive Rehabilitation in Older Adults" - Poster presentation, Biomedical Engineering Conference 2026, HKIE Biomedical Division; Champion, Best Poster Award (Undergraduate Category).

Honours & Awards

- Champion, Best Poster Award (Undergraduate Category), Biomedical Engineering Conference 2026, HKIE Biomedical Division
- Mitacs Globalink Research Internship Award, 2025
- Commercial Radio 50th Anniversary Scholarship, 2025
- Link University Scholarship, 2023